

CHAPTER 1

Introduction to Project Management

DISCUSSION QUESTIONS

1. Why is there a new or renewed interest in the field of project management?

Answer :

More and more projects are being done by a variety of organizations. The projects are more complex and often involve the use of new technologies. Organizations are struggling to find better ways to manage their projects.

2. What is a project, and what are its main attributes? How is a project different from what most people do in their day-to-day jobs? What is the triple constraint? What other factors affect a project?

Answer :

A project is “a temporary endeavor undertaken to create a unique product, service, or result” (PMBOK Guide, 2013). In addition to being temporary and unique, other attributes of projects are that they are developed using progressive elaboration, require resources from various areas, should have a primary customer or sponsor, and involve uncertainty. Projects are different from day-to-day activities primarily because they have focused goals and definite beginning and ending dates. The triple constraint is managing scope, time, and cost goals. Other factors that affect a project include quality, risk, human resources, communications, and stakeholders.

3. What is project management? Briefly describe the project management framework, providing examples of stakeholders, knowledge areas, tools and techniques, and project success factors.

Answer :

Project management is “the application of knowledge, skills, tools, and techniques to project activities in order to meet project requirements” (PMBOK Guide, 2013). The project management framework graphically shows the process of beginning with stakeholders’ needs and expectations, applying the nine project management knowledge areas and various tools and techniques to lead to project success and then enterprise success. For example, if a project were to implement an ERP system for a large company, the stakeholders would include managers and users from many different departments (finance, manufacturing, human resources, IT, etc.), all nine knowledge areas would be important, numerous tools and techniques would be applied (see Table 1-1), and project success might be based on implement key functions by a certain time for a certain cost or having the new system pay for itself within a certain time period.

4. What is a program? What is a project portfolio? Discuss the relationship between projects, programs, and portfolio management and the contributions they each make to enterprise success.

Answer :

A program is “a group of projects managed in a coordinated way to obtain benefits and control not available from managing them individually” (PMBOK Guide, 2013) Project portfolio management focuses on managing projects as

is as a portfolio of investments that contribute to the entire enterprise's success. Projects are part of programs which are part of portfolios.

5. What is the role of the project manager? What are suggested skills for all project managers and for IT project

managers? Why is leadership so important for project managers? How is the job market for IT project managers?

Answer :

The project manager is ultimately responsible for project success. Many suggested skills are listed in this chapter, including strong leadership skills, organizational skills, technical skills, and many soft skills. IT project managers require the same skills as general project managers, but they should also know something about the technology used for the project and the types of people who work on information technology projects. Leading by example is the most important trait of effective project managers. The job market for information technology project managers continues to remain strong, especially for those with strong business and leadership skills.

6. Briefly describe some key events in the history of project management. What role does the Project Management Institute and other professional societies play in helping the profession?

Answer :

Some people say that building the Egyptian pyramids or the Great Wall of China were projects, but modern project management began with the Manhattan Project or development of the atomic bomb. That project took about three years and cost almost \$2 billion in 1946 and had a separate project manager and technical manager. Gantt charts were first used in 1917, and network diagrams were used in 1958. PMI is the main professional society for project managers, and they run the PMP certification program.

7. What functions can you perform with project management software? What are the main differences between low-end, midrange, and high-end project management tools?

Answer :

Project management software can assist in developing schedules, communicating information, tracking progress, etc. Low-end tools are the least expensive, and several are available as apps. Midrange tools can usually create Gantt charts, perform critical path analysis, etc. High-end tools often perform portfolio management and can be used across a large organization.

8. Discuss ethical decision that project managers often face. Do you think a professional code of ethics makes it easier to work in an ethical manner?

Answers will vary. Some examples might include working on projects that you personally do not believe in (such as military projects, projects for different political parties, etc.), being asked to hire a friend or relative who is not as qualified as someone else, being offered bribes, etc. It is still difficult to work in an ethical manner even with a professional code of ethics.

CHAPTER 2

The Project Management and Information Technology Context

DISCUSSION QUESTIONS

1. **What does it mean to take a systems view of a project? How does taking a systems view of a project apply to project management?**

Taking a systems view means looking at the big picture of how a particular project fits into the rest of the organization. It is important for project managers to understand the broader organizational environment to ensure their projects meet organizational needs.

2. **Explain the four frames of organizations. How can they help project managers understand the organizational context for their projects?**

The four frames of organizations are summarized below:

Structural: deal with how the organization is structured and focus on roles and responsibilities. It's important to understand these roles and responsibilities when dealing with project stakeholders, especially in procuring resources.

Human resources: focuses on meeting the needs of the organization and its people. Project managers must understand various human resources policies and procedures.

Political: addresses organizational and personal politics. Many project managers fail because they do not understand the political environment.

Symbolic: focuses on symbols and meanings. It's important to understand an organization's culture, dress code, work ethic, and so on in managing projects.

3. **Briefly explain the differences between functional, matrix, and project organizations. Describe how each structure affects the management of the project.**

Functional organizations have managers or vice presidents in specialties such as engineering, manufacturing, information technology, and so on. Their staffs have specialized skills in their respective disciplines. Project organizations have project managers instead of functional managers reporting to the CEO. Matrix organizations represent the middle ground between functional and project structures. Personnel often report to both a functional manager and one or more project managers. Project managers have the most authority in project organizational structures followed by matrix, and then functional.

4. **Describe how organizational culture is related to project management. What type of culture promotes a strong project environment?**

Organizational culture is a set of shared assumptions, values, and behaviors that characterize the functioning of an organization. This culture can definitely impact project management. For example, if an organization values project management and follows the guidelines for applying it, it will be much easier to practice good project management. Project work is most successful in an organizational culture where employees identify more with the organization, where work activities emphasize groups, and where there is strong unit integration, high risk tolerance, performance-based rewards, high conflict tolerance, an open-systems focus, and a balanced focus on people, control, and means-orientation.

5. Discuss the importance of top management commitment and the development of standards for successful project management. Provide examples to illustrate the importance of these items based on your experience on any type of project.

Top management commitment is the number one factor associated with the success of information technology projects, so it's very important to get and maintain this support. Top management can help project managers get adequate resources, approve unique project needs, get cooperation from other parts of the organization, and provide support as a mentor and coach to project managers. Examples will vary.

6. What are the phases in a traditional project life cycle? How does a project life cycle differ from a product life cycle? Why does a project manager need to understand both?

A traditional project life cycle is a collection of project phases concept, development, implementation, and close-out. These phases do not vary by project. Product life cycles vary tremendously based on the nature of the project. For example, the Systems Development Life Cycle (SDLC) could follow the waterfall model, spiral model, incremental release model, prototyping model, or RAD model. Using the general phases of the SDLC (information systems planning, analysis, design, implementation, and support) there could be a project to develop a strategic information systems plan; another project to complete a systems analysis for a new system; another project to create a detailed database design; another to install new hardware or software; and another to provide new user training.

7. What makes IT projects different from other types of projects? How should project managers adjust to these differences?

IT projects are different from other types of projects because they can be very diverse in terms of size and complexity, they often include team members with very diverse backgrounds and skills, and the technologies involved are also very diverse. Project managers should adjust to these differences by paying careful attention to the goals of the project and the needs of various stakeholders.

8. Define globalization, outsourcing, virtual teams, and agile project management, and describe how these trends are changing IT project management.

Globalization has created a "flat" world where everyone is connected and the "playing field" is level for many more participants. Outsourcing is when an organization acquires goods and/or services from an outside source. Agile project management is a method for managing projects when requirements are unclear or change quickly. Virtual teams occur when a group of individuals who work across time and space using communication technologies. Each of these trends has affected the way in which project work is done and how projects need to be managed. It is very rare for a project team to sit in the same work area and work at the same time. Management and coordination is much more complicated.

CHAPTER 3

The Project Management Process Groups: A Case Study

DISCUSSION QUESTIONS

1. Briefly describe what happens in each of the five project management process groups (initiating, planning, executing, monitoring and controlling, and closing). What types of activities occur before initiating a project? Descriptions of what happens in each of the five process groups are listed below:

Initiating includes actions to commit to begin or end projects and project phases. Some deliverables include defining the business need for the project, getting a project sponsor, and selecting a project manager.

Planning includes creating workable plans for the entire project. Every knowledge area requires development of some plans.

Executing involves coordinating everyone and everything to carry out the project plans. Deliverables include the products of the project or phase.

Monitoring and controlling ensures the project objectives are met. Deliverables include monitoring and measuring progress and taking corrective action.

Closing involves bringing the project to an orderly end. Deliverables include organizational process assets updates and closed procurements.

Before initiating a project, organizations often determine high-level scope, time, and cost constraints for a project, identify the project sponsor, select the project manager, and develop a business case for the project, to name a few.

2. Approximately how much time do good project managers spend on each process group and why.

Andy Crowe's research found that alpha project managers spend about 2% of their time initiating, 21% planning, 69% executing, 5% monitoring and controlling, and 3% closing projects. The most time should be spent on executing, followed by planning.

3. Why do organizations need to tailor project management concepts, such as those found in the PMBOK® Guide, to create their own methodologies?

Organizations have different ways of doing things, so they often develop their own methodologies.

4. What are some of the key outputs of each process group?

See the tables for each process group. Examples might include a charter, a project management plan, deliverables, performance reports, and closed procurements.

5. What are some of the typical challenges project teams face during each of the five process groups? You can frame your discussion based on a project described in one of the feature boxes in this chapter (for example, the What Went Right? or What Went Wrong? feature). You can also frame your discussion on one of PMI's Project of the Year Award winners, or on a well-known project failure like the Denver International Airport baggage handling system.

Answers will vary. Suggestions might include setting priorities during initiating, getting stakeholder inputs during planning, meeting schedule deadlines during execution, collecting timely data during monitoring and controlling, and defining good procedures during project closing.

6. What are the main differences between the two versions of the JWD Consulting case study? When should you use a more prescriptive or agile approach? Do you think users of the JWD Consulting Intranet site would prefer one release of the software or several incremental ones? What are some pros and cons of each approach?

The main difference is that three iterations of software are delivered instead of one. An agile approach normally works best when requirements are unclear or change frequently. Answers will vary on user preferences. There could be training issues with multiple releases, but there could be business value in getting some releases earlier.

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CHAPTER 4

Project Integration Management

DISCUSSION QUESTIONS

1. **Describe project integration management in your own words. How does project integration management relate to the project life cycle, stakeholders, and the other project management knowledge areas?**

Project integration management means tying together all of the other aspects involved in a project to make it a success. Integration management relates to the project life cycle in that it is done in all of the project life cycle phases. As the project progresses, integration management becomes more focused. Integration management relates to stakeholders because it requires the project manager to know all of the project stakeholders, to know their interests and concerns about the project, and to manage relationships with them. Integration management pulls together information from all of the other knowledge areas.

2. **Describe options that organizations have for selecting projects that align with their mission or strategy, and describe how each might work differently in the selection of IT projects.**

Many organizations use a SWOT analysis, which involves identifying and evaluating an organization's strengths, weaknesses, opportunities, and threats. After identifying strategic goals, the next step in the planning process for selecting information technology projects is to perform a business area analysis. This analysis outlines business processes that are central to achieving strategic goals and helps determine which ones could most benefit from information technology. Then, the next step is to start defining potential information technology projects, their scope, benefits, and constraints. The last step in the planning process for selecting information technology projects is choosing which projects to do and assigning resources for working on them. . Many organizations also use financial methods to select projects, such as ROI, payback, and NPV. Answers will vary on the second part of the question. All of the project selection methods are used.

3. **Summarize key work involved in each of the six processes for project integration management.**

Project integration management includes the following processes:

Developing the project charter involves working with stakeholders to create the document that formally authorizes a project. Project charters can have different formats, but they should include basic project information and signatures of key stakeholders.

Developing the project management plan involves coordinating all planning efforts to create a consistent, coherent document—the project management plan. The main purpose of project plans is to facilitate action.

Directing and managing project work involves carrying out the project plans by performing the activities included in it. Project plan execution should require the majority of a project's budget.

Monitoring and controlling project work is needed to meet the performance objectives of the project. The project team should continuously monitor project performance to assess the overall health of the project.

Performing integrated change control involves coordinating changes that affect the project's deliverables and organizational process assets. A change control system often includes a change control board (CCB), configuration management, and a process for communicating changes.

Closing the project or phase involves finalizing all project or phase activities. It is important to follow good procedures to ensure that all project activities are completed and that the project sponsor accepts delivery of the final products, services, or results of the project.

4. Either from your own experience or by searching the Internet, describe a well-planned and executed project. Describe a failed project. What elements of project integration might have contributed to the success or failure of each?

Answers will vary. Several examples are provided in the text, especially in the What Went Right and Wrong sections.

5. Discuss the importance of following a well-integrated change control process on IT projects. What consequences can result from not following these best practices? What types of change control would be appropriate for small IT projects? What types of change control would be appropriate for large ones?

Answers will vary. One answer might be that it is important to follow a good integrated change control process on IT projects to avoid scope creep, incompatibility problems, and to make effective use of resources and new technologies. Additional suggestions for managing integrated change control might include locating key project stakeholders, assigning users to project teams, and providing team-based incentives for effectively managing project changes. Larger projects obviously need more change control than smaller ones. Larger projects would probably have formal change control forms and a change control board, while a smaller project might need less formal processes, such as just the project manager's or sponsor's approval in a verbal or written form.

